PUSL3190 Computing Project

Project Initiation Document (PID)

Smart Financial Tracker: A Web-Based Personal Finance Tracking and Analysis System

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Contents

Chapter 01 – Introduction 3
1.1 Background and Context 3
1.2 Problem Statement 3
1.3 Scope and limitations 4
Project Scope 4
Project Limitations 4
1.4 Expected Impact and Stakeholders 5
Chapter 02 – Business Case 6
2.1 Business Need 6
2.2 Business Objectives 6
Chapter 03 – Project Objectives 6
3.1 Overall Project Objective 6
3.2 Project Objectives 6
Chapter 04 – Literature Review 7
4.1 Personal Financial Management Systems 7
4.2 Budgeting and Financial Awareness 7
4.3 Data Visualisation in Financial Applications 8
4.4 Identified Research, Technological, and Methodological Gaps 9
Research Gap: Limited Support for Behavioural Financial Change 9
Technological Gap: Fragmented Architectures and Delayed Feedback 9
Methodological Gap: Insufficient User-Centred Development Practices 9
Chapter 05 Method of Approach 10
5.1 Research Design 10
5.2 Data Sources and Data Handling 10
5.3 Tools, Technologies, and Frameworks 10
5.4 System Logic and Processing 10
5.5 Evaluation and Validation Strategy 10
5.6 Project Management Methodology 10
Chapter 06 – Conceptual Diagram 11
Chapter 07 – Initial Project Plan 12
Chapter 08 – Risk Analysis 13
References 14
Appendix 16

Chapter 01 – Introduction

1.1 Background and Context

Personal financial management has become an essential competency within the modern digital economy. Individuals are increasingly responsible for managing multiple income sources, recurring expenses, digital subscriptions, and short-term financial commitments. The widespread adoption of cashless transactions and online payment platforms has increased convenience; however, it has also reduced individuals' ability to accurately monitor spending behaviour (OECD, 2019). As a result, limited financial visibility has often led to ineffective budgeting practices and increased financial stress.

Despite the availability of numerous digital finance applications, many individuals continue to rely on manual financial tracking methods, including notebooks and spreadsheets. These approaches are time-consuming, prone to human error, and provide limited analytical insight into spending patterns. Existing digital solutions frequently prioritise advanced automation or bank integrations rather than usability, which has resulted in reduced long-term adoption due to complexity (Laudon & Laudon, 2020).

The Smart Financial Tracker (SFT) project proposes a web-based personal finance tracking system that prioritises simplicity, clarity, and user awareness. The system enables users to manually record income and expense transactions, monitor weekly and monthly budgets, and visualise financial information through dashboards and charts. The project is developed as part of an academic requirement and demonstrates the practical application of computer science principles to address a real-world financial management problem.

1.2 Problem Statement

Personal financial management continues to be a challenge for many individuals due to limited visibility and control over daily financial activities. The key problems identified are as follows:

- Many individuals lack a structured and consistent method for recording income and expenses, resulting in incomplete or inaccurate financial records.
- Manual tracking methods such as notebooks and spreadsheets are time-consuming, prone to human error, and difficult to maintain over long periods.
- Users often fail to recognise spending patterns early due to the absence of real-time summaries and visual feedback.
- Existing digital finance applications frequently prioritise automation and complex features, which discourages continued usage by non-technical users.
- The lack of timely financial insight contributes to overspending, ineffective budgeting, and delayed financial decision-making.
- The Smart Financial Tracker aims to address these issues by providing a simple, structured, and insight-driven platform for personal finance monitoring.

1.3 Scope and limitations

Project Scope

- **The system allows users to manually record income and expense transactions in a structured manner.**
  This approach ensures consistent and accurate data entry while encouraging users to actively engage with their financial activities. Manual recording also improves awareness of spending habits and reduces reliance on automated processes that may obscure financial behaviour.
- **Users can categorise financial records to better organise and analyse spending behaviour.**
  Categorisation supports clearer interpretation of financial data by grouping similar expenses and income sources. This enables users to compare spending across categories and identify areas where financial adjustments may be required.
- **Weekly and monthly budgets can be set and monitored to support spending control.**
  Supporting both weekly and monthly budgeting allows users to manage finances across different time horizons. This structure helps users detect overspending early and adjust behaviour before long-term financial imbalance occurs.
- **Real-time financial summaries and visual dashboards provide clear insight into income, expenses, and balances.**
  Visual dashboards present financial information in an intuitive format that supports quick understanding and pattern recognition. Real-time updates ensure that users base decisions on the most current financial data available.
- **Secure user authentication ensures that financial data is stored and accessed safely.**
  Authentication mechanisms restrict access to authorised users only, protecting sensitive personal financial information. This also increases user trust in the system and supports responsible data management practices.

Project Limitations

- The system does not integrate with banks or external financial services for automatic transaction imports.
- Advanced financial features such as investments, loans, and credit management are not included.
- The system stores financial records using a single base currency; however, currency conversion may be provided as a reference feature without supporting multi-currency transaction storage.
- The system is designed for individual users and does not support organisational or multi-user finance management.
- These limitations are applied to maintain simplicity and ensure completion within the academic timeframe.

1.4 Expected Impact and Stakeholders

The Smart Financial Tracker is expected to improve personal financial awareness by enabling users to clearly monitor income, expenses, and budgeting performance through visual and structured feedback.
Key Stakeholders:

1. **End Users – Individuals who use the system to track and manage personal finances.**
   End users benefit directly from improved financial visibility and structured budgeting tools. The system supports better understanding of spending patterns and encourages responsible financial behaviour.
2. **Project Developer – Responsible for system design, implementation, testing, and documentation.**
   The project developer applies software engineering principles to design, build, and validate the system. This role is also responsible for ensuring system quality, security, and alignment with academic requirements.
3. **Project Supervisor – Provides academic guidance, feedback, and evaluation.**
   The supervisor ensures that the project meets academic standards and learning outcomes. Guidance is provided throughout the project lifecycle to support technical correctness and research quality.
4. **Academic Institution (NSBM / Plymouth University) – Oversees assessment quality and learning outcomes.**
   The academic institution ensures that the project adheres to curriculum requirements and assessment regulations. It also evaluates the project's contribution to the student's academic development.
5. **Future Developers or Maintainers – Potential contributors responsible for system enhancement or maintenance.**
   Future developers may extend or maintain the system after project completion. Clear documentation and modular design support maintainability and future feature enhancements.

Chapter 02 – Business Case

2.1 Business Need

Many individuals experience difficulty in managing personal finances due to the lack of simple, structured, and insightful financial management tools. Manual tracking approaches are inefficient and error-prone, while many existing digital applications provide limited budget feedback or impose unnecessary complexity (Xiao & O'Neill, 2018).

The Smart Financial Tracker addresses this need by offering a centralised web-based platform for recording income and expenses, monitoring budgets, and visualising financial data. The system is intended to support everyday financial decision-making by improving clarity and awareness of personal spending behaviour.

2.2 Business Objectives

- To provide a structured platform for personal financial tracking.
- To support effective budgeting and spending control.
- To improve financial awareness through visual insights.
- To enhance user confidence in financial decision-making.
- To reduce the complexity associated with existing financial tools.

Chapter 03 – Project Objectives

3.1 Overall Project Objective

The objective of this project is to design and implement a secure and scalable web-based personal finance management system using the MERN technology stack. The system is intended to support effective tracking, categorisation, and analysis of individual financial activities through real-time visual feedback and intelligent insights, thereby improving users' financial awareness and decision-making ability.

3.2 Project Objectives

- To develop a web application that enables users to create, view, update, and delete income and expense records in a structured and reliable manner.
- To implement a rule-based transaction categorisation mechanism capable of accurately classifying the majority of recurring and previously recorded financial transactions.
- To utilise the MERN stack throughout the development process while adhering to established software engineering practices and completing the system within the allocated academic timeframe.
- To present financial information through personalised dashboards and visual representations that support informed financial decisions and enhance user confidence.
- To complete system development, testing, and documentation within the defined project schedule, ensuring readiness for final academic evaluation by March 2026.

Chapter 04 – Literature Review

4.1 Personal Financial Management Systems

Personal Financial Management (PFM) systems are widely recognised as digital tools that support individuals in monitoring income, expenses, and savings while improving overall financial awareness. Research demonstrates that structured financial tracking contributes positively to budgeting discipline and financial capability by encouraging consistent engagement with personal financial data (Xiao & O'Neill, 2018). Such engagement has been shown to improve users' understanding of spending patterns and financial outcomes over time.

Despite these benefits, sustained adoption of PFM systems remains inconsistent. Evidence indicates that long-term usage is strongly influenced by system usability rather than the availability of advanced features. Studies examining user behaviour reveal that applications perceived as complex are frequently abandoned, even when they offer sophisticated analytical capabilities (Kim, et al., 2003). This highlights a limitation in systems that prioritise functionality without sufficient consideration of user experience.

In recent financial technologies, automation—particularly through bank integrations and transaction synchronisation—has become increasingly prominent. While automation reduces manual data entry and improves data accuracy, it also introduces additional layers of complexity and dependency on external services (Laudon & Laudon, 2020). These characteristics may present barriers for users with limited technical literacy and raise concerns related to transparency and data privacy.

Conversely, evidence from behavioural finance literature indicates that manual or semi-manual transaction recording can enhance financial awareness. Active involvement in entering financial data has been associated with greater cognitive engagement and stronger accountability for spending decisions (Baker & Yannelis, 2017). However, such systems receive limited attention in contemporary fintech research, which continues to favour automation-driven approaches (Xiao & O'Neill, 2018).

Taken together, existing literature reveals an imbalance between efficiency-oriented automation and user-centred financial awareness. While automated systems dominate both research and practice, simplified financial tracking tools that emphasise behavioural engagement and long-term usability remain underexplored.

4.2 Budgeting and Financial Awareness

Budgeting is consistently recognised as a central mechanism in effective personal financial management. Empirical research demonstrates that individuals who actively monitor budgets are more likely to regulate discretionary spending, reduce financial stress, and improve overall financial stability (OECD, 2019).Budgeting practices that encourage regular review and reflection have been shown to support more informed financial decision-making.

Despite this, the effectiveness of budgeting tools is strongly influenced by how financial information is presented and updated. Evidence suggests that budget awareness is

significantly enhanced when users receive timely and ongoing feedback regarding spending progress rather than retrospective summaries (OECD, 2019). Systems that rely solely on end-of-period reports often fail to support behavioural adjustment during the budgeting cycle.

A further limitation identified in existing applications is the predominant focus on long-term or monthly budgeting structures. Research indicates that many users make frequent, short-term spending decisions that require immediate feedback to support behavioural regulation (Xiao & O'Neill, 2018)Budgeting tools that do not accommodate shorter budgeting intervals may therefore offer limited practical value for everyday financial management.

Behavioural finance literature further highlights that awareness alone is insufficient to influence financial behaviour. The integration of behavioural reinforcement mechanisms, such as progress indicators and reminders, has been shown to support behavioural adjustment and sustained engagement (Thaler & Sunstein, 2008).However, many existing financial applications provide limited behavioural feedback, reducing their capacity to translate budgeting information into meaningful action.

Overall, the literature suggests a gap between established budgeting theory and its practical implementation in digital financial systems, particularly in relation to short-term feedback and behavioural reinforcement.

4.3 Data Visualisation in Financial Applications

Data visualisation is widely acknowledged as a critical component of effective financial information systems. Visual representations such as charts and graphs support users in interpreting complex financial data by reducing cognitive effort and enabling pattern recognition (Few, 2013). In personal finance contexts, well-designed visualisations allow users to quickly understand spending distributions, income trends, and budget performance.

Research emphasises that effective financial visualisation prioritises clarity and relevance over visual complexity. Studies on information design demonstrate that minimal, purpose-driven dashboards enhance comprehension and decision-making by focusing attention on key metrics (Tufte, 2001).Visual elements that directly support user goals are more likely to facilitate meaningful insight than dense or decorative interfaces.

However, not all financial applications adhere to these principles. Evidence indicates that excessive visual elements, poor visual hierarchy, and information overload can overwhelm users and obscure critical insights (Few, 2013) (Ware, 2013). Dashboards that prioritise aesthetics over analytical clarity may reduce usability and discourage sustained engagement.

In many commercial financial systems, data visualisation is treated as an optional enhancement rather than a core analytical tool. This approach contrasts with established research, which positions visualisation as a fundamental mechanism for supporting understanding and behavioural awareness. The limited application of evidence-based visual design principles highlights an underutilisation of visualisation as a behavioural support feature in personal finance applications.

4.4 Identified Research, Technological, and Methodological Gaps

A synthesis of existing literature reveals several interconnected gaps that constrain the effectiveness of current personal financial management solutions. These gaps span behavioural support, technological design, and development methodology, and collectively inform the rationale for this project.

Research Gap: Limited Support for Behavioural Financial Change

While many financial systems effectively record and summarise transactions, they offer limited support for behavioural reflection and habit formation. Research indicates that financial awareness does not consistently translate into behavioural change unless users are encouraged to understand the underlying causes of their spending patterns (Baker & Yannelis, 2017).Most systems present historical data without sufficient contextual interpretation, reducing their influence on future decision-making.

Additionally, behavioural studies highlight the importance of reinforcement mechanisms in shaping financial behaviour. Systems that lack behavioural cues, such as reminders or progress indicators, often fail to convert financial information into actionable insight (Thaler & Sunstein, 2008). This reveals a gap in solutions that actively support behavioural improvement rather than passive monitoring.

Technological Gap: Fragmented Architectures and Delayed Feedback

From a technological perspective, many personal finance systems rely on fragmented architectures that separate data processing from presentation layers. Such designs frequently result in delayed updates and reduced system responsiveness, limiting real-time financial awareness (Laudon & Laudon, 2020).Applications that depend on periodic data synchronisation further restrict the availability of immediate financial feedback, particularly for short-term budgeting and expense monitoring.

There is limited evidence of lightweight, unified architectures that prioritise real-time feedback while maintaining simplicity and maintainability. This technological gap restricts the ability of existing systems to provide timely, insight-driven financial information.

Methodological Gap: Insufficient User-Centred Development Practices

The literature also highlights methodological limitations in the development of financial software systems. Rigid development approaches often prioritise technical completeness over usability, resulting in systems that fail to align with real-world user needs (Sommerville, 2016). Limited user involvement during development contributes to reduced adoption and early abandonment.

There is a lack of documented approaches that integrate iterative development practices with user-centred financial design. This methodological gap underscores the need for development processes that continuously align system functionality with user behaviour and expectations.

Chapter 05 Method of Approach

5.1 Research Design

A prototype-based development approach grounded in applied computer science practice is adopted for the Smart Financial Tracker project. This approach supports iterative refinement of system features, interface design, and business logic through continuous testing and evaluation.

5.2 Data Sources and Data Handling

User-generated financial data serve as the primary data source. Transactions are manually entered and securely stored in a database associated with authenticated user accounts. External financial services are not integrated, reducing data privacy risks.

5.3 Tools, Technologies, and Frameworks

The system is developed using a full-stack technology stack. The frontend is implemented using React, while the backend is developed using Node.js and Express. MongoDB is used for data storage. Git and GitHub support version control and source code management.

5.4 System Logic and Processing

The system focuses on deterministic financial calculations, including income totals, expense totals, balances, budget usage, and time-based comparisons. Calculations are performed dynamically to ensure immediate updates.

5.5 Evaluation and Validation Strategy

Functional validation and manual testing are used to verify data accuracy, system stability, and budget logic. System behaviour is evaluated across multiple usage scenarios.

5.6 Project Management Methodology

An Agile-inspired iterative development approach is applied. Development tasks are organised into manageable iterations, supported by frequent testing and regular version control commits.

Chapter 06 – Conceptual Diagram

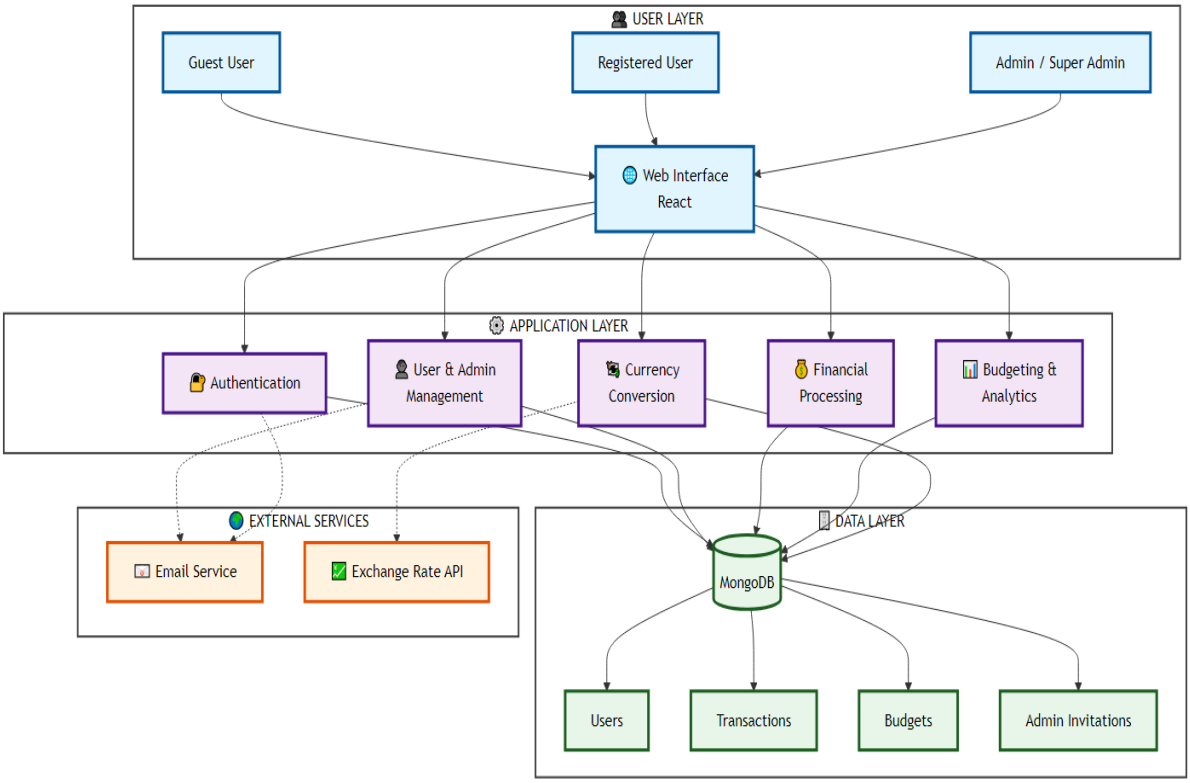

Figure 1: Layered System Architecture Diagram of the Smart Financial Tracker

Chapter 07 – Initial Project Plan

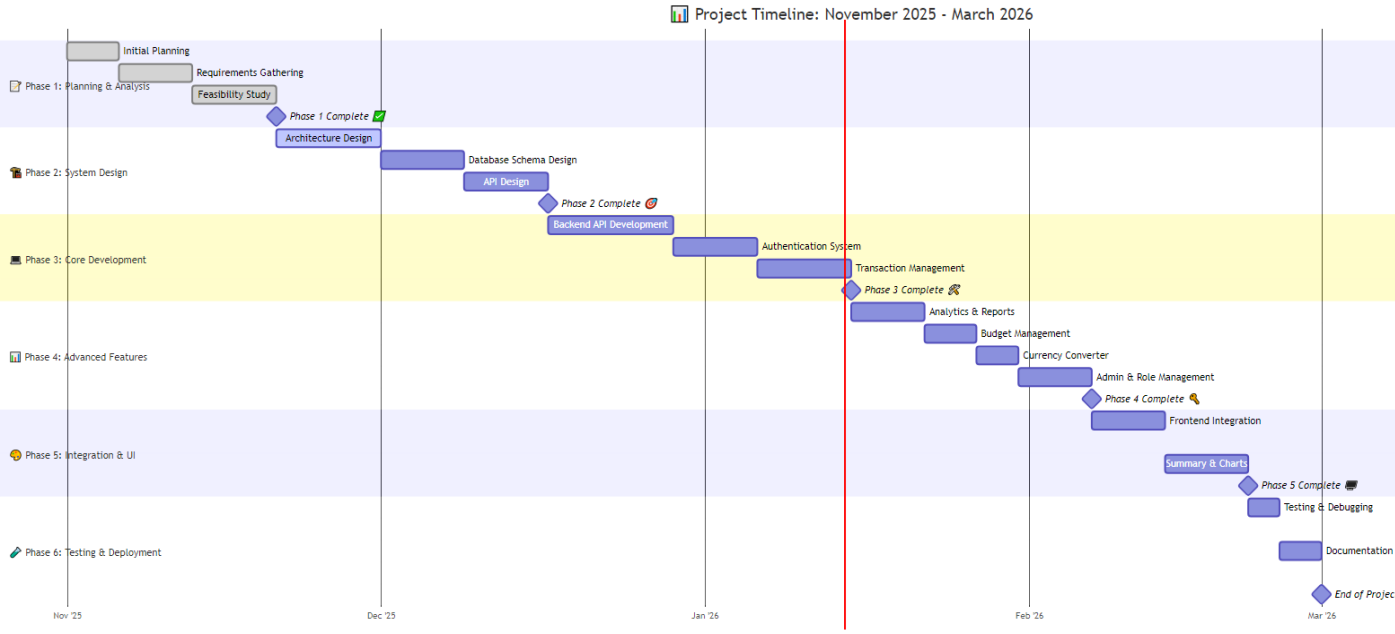

Figure 2: Project Development Timeline (Gantt chart)

Chapter 08 – Risk Analysis

Budget constraints.
As the project is developed within an academic context, financial resources are limited and rely mainly on free or open-source tools, which may restrict advanced deployment options. This risk is minimised by using open-source technologies and free-tier cloud services.

Technical complexity of the MERN stack.
Integrating frontend, backend, and database components may introduce development challenges and integration errors. This risk is managed through modular design, clear API structures, and incremental testing.

Performance issues in real-time data visualisation.
Frequent updates to dashboards and charts may reduce system responsiveness if not optimised. Efficient database queries and controlled UI re-rendering will be applied to maintain acceptable performance.

Errors in financial calculations.
Incorrect income or expense calculations could affect data accuracy and user trust. Validation logic and thorough testing across different financial scenarios will be used to reduce this risk.

Data security and privacy risks.
The system stores sensitive personal financial information, making it vulnerable to unauthorised access if security is insufficient. Secure authentication, encrypted password storage, and access controls will be implemented to protect user data.

Data loss or system failure.
Unexpected system or database failures could result in loss of financial records. Reliable data handling practices and error-handling mechanisms will be used to ensure data integrity.

Limited development time.
Academic deadlines and other coursework may limit the available time for development and testing. This risk is mitigated by prioritising core features and following an iterative development approach.

Scope creep during development.
Adding unplanned features may delay project completion and reduce overall quality. Strict adherence to the defined project scope and regular progress reviews will prevent unnecessary expansion.

Usability challenges.
If the interface is not intuitive, users may find the system difficult to use and discontinue usage. User-centred design principles and iterative interface improvements will be applied to enhance usability.

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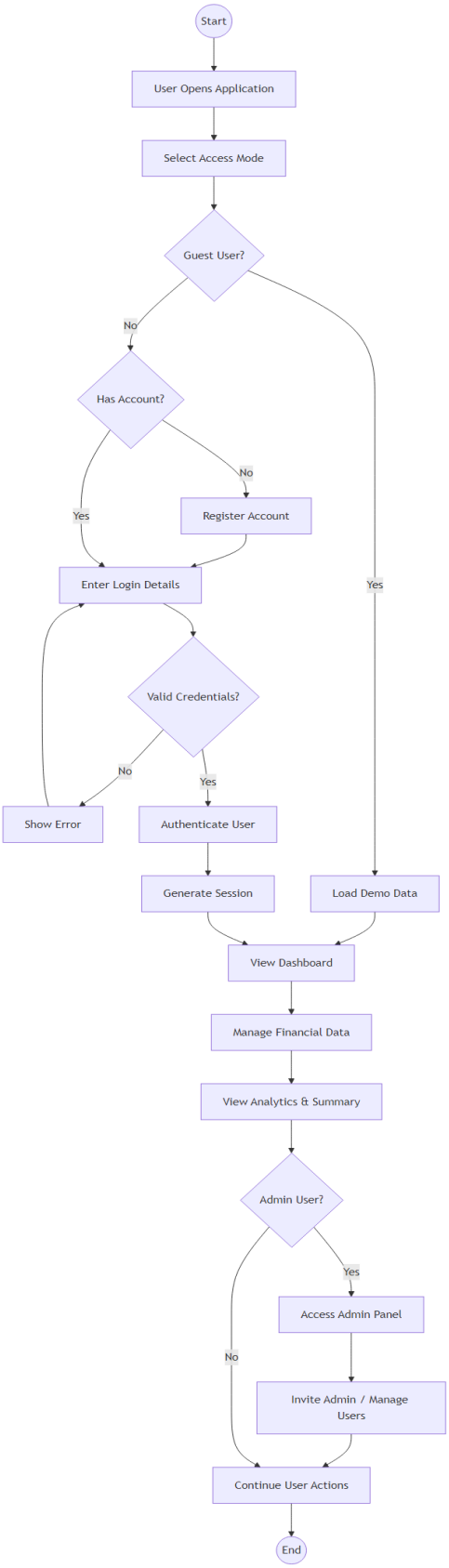

Figure 3: User Authentication and System Interaction Flow Diagram

Appendix

Area of Analysis	Existing Approaches	Identified Gap	Project Contribution
Financial Tracking Approach	Many existing systems prioritise automated transaction syncing and integration with external financial services to reduce manual input.	Automation-focused designs can reduce user engagement and limit behavioural awareness, particularly for non-technical users.	The system emphasises manual and semi-manual transaction entry to encourage active user involvement and financial awareness.
Behavioural Financial Awareness	Financial applications commonly focus on recording and summarising historical transaction data.	Limited support is provided for behavioural reflection or habit improvement based on spending patterns.	Interactive dashboards and contextual summaries are used to support behavioural understanding and informed decision-making.
Budgeting Support	Most tools support monthly or long-term budgeting structures.	Short-term budgeting needs, such as weekly spending control, are insufficiently addressed.	The system incorporates both weekly and monthly budgeting with continuous monitoring.
Data Visualisation	Financial data is often presented using dense or visually complex dashboards.	Excessive visual complexity can overwhelm users and reduce clarity.	Minimal, purpose-driven visualisations are adopted to enhance clarity and insight.
System Responsiveness	Many systems rely on fragmented architectures and periodic data updates.	Delayed feedback limits real-time financial awareness and timely decision-making.	A unified system architecture enables near real-time updates and consistent data flow.
Development Methodology	Traditional, rigid development approaches are commonly applied.	Limited user alignment and late usability feedback reduce system adoption.	An iterative, Agile-inspired development approach supports continuous refinement and user alignment.